MQTT Firewall Intrusion Detection Report

Generated: 2026-04-30 00:10:12
Source file: `JSONL (1,573 verdicts) + .log (1,573 verdicts)`
Time range: 00:09:41 → 00:09:57

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1. Executive Summary

The LightGBM-based MQTT firewall processed **15,696 packets** (17,276 log lines) and issued **1,574 flow-level verdicts**. Of these, **1,568 (99.6%) were classified as attacks** and **6 (0.4%) as benign**. Mean model confidence on attack flows was **0.9464**.

Metric	Value
Total log lines	17,276
Total packets intercepted	15,696
Total flow verdicts	1,574
Attack verdicts	1,568 (99.6%)
Benign verdicts	6 (0.4%)
Observation window	00:09:41 → 00:09:57

2. Attack Type Breakdown

Attack Type	Verdict Count	Share (%)
MQTT-DoS-Publish_Flood	1,540	97.8%
Recon-OS_Scan	28	1.8%
benign	6	0.4%

3. Model Confidence Statistics

Statistics computed over **attack-classified flows only**.

Statistic	Value
Sample size	1,568 flows
Minimum confidence	0.8029
Maximum confidence	1.0000

Statistic	Value
Mean confidence	0.9464
Flows with confidence ≥ 0.90	1,567 (99.9%)

4. Flow Statistics

Metric	Min	Max	Mean
Packets per flow	3	10	10.0
Bytes per flow	3,072	10,240	10,215
Flow duration (s)	0.0002	0.3007	0.0336

5. Top Attacker Flows

19 distinct `(src_ip:src_port, attack_type)` flows detected.

Source (ip:port)	Classified Attack Type	Verdict Count
127.0.0.1:34150	MQTT-DoS-Publish_Flood	191
127.0.0.1:48133	MQTT-DoS-Publish_Flood	186
127.0.0.1:50525	MQTT-DoS-Publish_Flood	173
127.0.0.1:34043	MQTT-DoS-Publish_Flood	169
127.0.0.1:52725	MQTT-DoS-Publish_Flood	153
127.0.0.1:43293	MQTT-DoS-Publish_Flood	148
127.0.0.1:53089	MQTT-DoS-Publish_Flood	138
127.0.0.1:45091	MQTT-DoS-Publish_Flood	133
127.0.0.1:38231	MQTT-DoS-Publish_Flood	125
127.0.0.1:52696	MQTT-DoS-Publish_Flood	124
127.0.0.1:53089	Recon-OS_Scan	5
127.0.0.1:52725	Recon-OS_Scan	4
127.0.0.1:45091	Recon-OS_Scan	4
127.0.0.1:38231	Recon-OS_Scan	3
127.0.0.1:43293	Recon-OS_Scan	3
127.0.0.1:34043	Recon-OS_Scan	3
127.0.0.1:34150	Recon-OS_Scan	3

Source (ip:port)	Classified Attack Type	Verdict Count
127.0.0.1:50525	Recon-OS_Scan	2
127.0.0.1:48133	Recon-OS_Scan	1

6. Last 10 Verdicts

Time	Verdict	Attack Type	Confidence	Source	Destination	Pkts	Bytes
00:09:47	ATTACK	MQTT-DoS-Publish_Flood	0.9418	127.0.0.1:34150	127.0.0.1:1883	10	10,240
00:09:47	ATTACK	MQTT-DoS-Publish_Flood	0.9392	127.0.0.1:43293	127.0.0.1:1883	10	10,240
00:09:57	ATTACK	MQTT-DoS-Publish_Flood	0.9140	127.0.0.1:53089	127.0.0.1:1883	9	9,216
00:09:57	benign	benign	0.0096	127.0.0.1:38231	127.0.0.1:1883	5	5,120
00:09:57	benign	benign	0.5976	127.0.0.1:34043	127.0.0.1:1883	6	6,144
00:09:57	ATTACK	MQTT-DoS-Publish_Flood	0.8029	127.0.0.1:48133	127.0.0.1:1883	7	7,168
00:09:57	benign	benign	0.0227	127.0.0.1:45091	127.0.0.1:1883	4	4,096
00:09:57	benign	benign	0.3204	127.0.0.1:52725	127.0.0.1:1883	3	3,072
00:09:57	benign	benign	0.0227	127.0.0.1:34150	127.0.0.1:1883	4	4,096
00:09:57	benign	benign	0.0209	127.0.0.1:43293	127.0.0.1:1883	4	4,096

Report generated by `scripts/parse_firewall_logs.py` — MQTT IDS/IPS Benchmarking Framework.